AppleSupport AI Agent Report

1. Aim :

This AI Agent is built for AppleSupport. I picked this brand because it had the most real data in the file we got, 13 real messages, mostly about the 2017 battery drain problem after an iPhone update.

Good means this : The bot should never promise something it cannot actually do on Twitter, like a refund or a repair. It should just understand the problem and ask the person to send a DM with the right details, the same way real AppleSupport agents did before.

Getting the escalation decision right matters more than writing a pretty reply. If we miss a fraud or safety case, that is much worse than sending an easy case to a human by mistake. So I made the system lean towards sending more things to humans when unsure.

Understanding the intent of a message is just a step to help us pick the right reply and the right past example. It is not the main goal by itself.

I chose not to build a few things. I did not make the bot write replies purely from imagination. Every reply is based on a real or realistic past reply, even when we use the optional AI rewrite through Groq. We also treat every message on its own, we do not track a full conversation history. I also did not build a separate mood detector, we just check for angry or urgent words directly in the escalation rule. No use of heavy AI models to find similar past messages, we used a simple word matching method that is easy to understand.

2. Data :

We built the data in three parts.

First, the 13 real messages. 8 of them are used to teach the bot how to reply. 5 of them are kept aside to test the bot fairly, the bot never saw these during training.

Second, 180 messages were written in the same style as real tweets. We needed these because 13 real ones are not enough to train anything, and also because 4 out of 6 topics we care about have zero real examples.

Third, 45 tricky examples written by hand to test hard situations, like fraud, safety, and cases that could confuse the escalation rule.

Every single test example is labeled with where it came from, real, written by us for training, or the tricky hand written ones. This way, nobody can hide behind one big score, we always know which part of the data made the number look good or bad.

3. How well it actually works, compared to simple methods :

We tested on 150 examples total, 100 written by us, 45 tricky ones, and 5 real ones, none of which the bot was trained on.

For guessing the topic of a message, a system that always guesses the same answer gets it right 17 percent of the time. A simple system that just looks for keywords gets it right 80 percent of the time.

Our actual trained model gets it right 77 percent of the time.

If we look at accuracy by where the data came from, it tells a clearer story. On the 5 real messages, we got 100 percent right, but that is too small a number to trust. On the 100 messages we wrote ourselves for testing, we got 86 percent right. On the 45 tricky ones, we got 56 percent right.

For deciding when to send a message to a human, if we always let the bot handle everything, it looks correct 85 percent of the time, but it would miss every single risky case. If we always send everything to a human, we catch all risky cases, but only 15 percent of decisions were needed to escalate in the first place, so this wastes a lot of human time. Our actual rule gets it right 67 percent of the time overall, but it catches 86 percent of the truly risky cases, which is the number that matters most for this brand.

For reply quality, our automatic scoring system rated the replies 4.28 out of 5 on average.

We also added an option to rewrite replies using a free AI model through Groq. This is turned off by default, and none of the numbers above depend on it being turned on.

4. Main problems found, with real examples :

Problem one. When someone says their phone got wet and will not turn on, the bot sometimes thinks it is a simple how to question, and gives instructions for setting something up, instead of understanding this is a broken phone.

Problem two. When someone gives a compliment, like thanking the support team, the bot sometimes reads it as a technical problem and replies with unrelated troubleshooting questions, or sends it to a human by mistake even though nothing is wrong.

Problem three. The overall topic accuracy number, 77 percent, hides a big gap. On easy normal messages we get 86 percent right, but on hard tricky messages we only get 56 percent right. One single number makes the system look better than it really is on the hard cases that matter most.

5. What we would do next

Get more real data, even a bigger sample from the same dataset, and run the same system again to see how much of our results change.

Test it against a bigger group of human ratings, done by more than one person, to check it is reliable.

Let the bot understand more than one topic in a single message, since right now it can only pick one topic even if a customer mixes two problems together.

Clean up old links and details from real historical replies before using them as examples, so the bot does not repeat outdated information.

Add a few fixed safe replies for dangerous situations, like a swelling battery or water damage, so the bot does not just pick the closest matching reply by word similarity in situations where safety matters.

Test different cutoff points for when the bot should send something to a human, based on what actually costs the business more, wasted human time or a missed risky case, instead of using one fixed number we picked ourselves.